10 – Reasoning effort (thinking)

AI Training for Beginners

Self-study training: use AI/LLMs safely in everyday life

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https://github.com/mgunthner/AI-Training

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10 – Reasoning effort (thinking)

Definition: In **thinking mode** (English: *Reasoning* or *Thinking*), the model produces intermediate steps before the actual answer and, so to speak, “thinks” before it answers.

> **Remember:** Thinking costs time and eats into your limit. Use it for tricky things, not for routine.

Key points

- **What happens?** The model works through a task in steps instead of writing straight away. What the mode is called differs from service to service.
- **When it pays off:** logic, math, planning with multiple constraints, complex code, multi-stage problems.
- **The price:** The answer comes slower and uses more tokens, and therefore more of your limit (topics 02, 15).
- **Not needed for:** rephrasing, translating, simple questions, collecting ideas.
- **Settings:** Some services have a switch in the chat, others offer levels (low, medium, high). In the API, there is often a budget for reasoning effort (topic 14). What your plan allows changes often: check the current details with your provider.
- **Visible thinking is no proof:** The displayed steps are no guaranteed truth and not necessarily the full reasoning. You check the final result too.
- **Rule of thumb:** Try standard mode first. If the result is superficial or contradictory, switch on thinking mode.

Example

Standard is enough:

```
Rephrase this sentence more friendly: "Send me the documents by
↪ tomorrow."
```

Thinking mode is worth it:

```
Plan my Saturday. Breakfast (60 min.), shopping (45 min.),
doctor's appointment at 11:00 a.m. (30 min.), visit to grandma (2
↪ hours).
Between appointments I need 20 minutes of travel time each.
I start at 8:00 a.m. and want to be done by 3:00 p.m. at the latest.
Create a schedule and tell me whether everything fits.
```

Check the schedule yourself afterwards.

Do / Don’t

Do	Don’t
Use thinking mode for multi-step tasks	Leave it on for every little thing
Spot-check the result and the calculation	Treat “it thought about it” as “it’s correct”
With a tight limit, test standard mode first	Stuff several tasks into one prompt

Safety note

The visible thinking steps can also reproduce data from your input. In thinking mode, don’t disclose more than you normally would (topic 04).

Mini checklist

- ☐ I know what thinking mode does.
- ☐ I know when it pays off and when it doesn’t.
- ☐ I know its price: time and limit.
- ☐ I also check results from thinking mode.

Exercise (5 min.)

Solve this logic puzzle once without and once with thinking mode. If your service doesn’t have one, use this for the second run: “Think step by step.”

```
Four colleagues (A, B, C, D) sit next to each other in four seats,
from left to right. D sits farthest left. A doesn't sit at the edge.
B sits directly to the left of C. Who sits where?
```

Compare speed, length, and result. Check: D, A, B, C.

Next

- Extra exercises: `02_extra_exercises/10_extra_exercises`
- Test: Test B (Good results)
- Related: 03 Model selection, 06 Prompt techniques, 15 Cost control

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